

ALGEBRA I - AUGUST-NOVEMBER 2024

Class timings: Tuesday - Thursday 2 : 00 – 3 : 30.

Tutorial timings: Friday 5:00 PM. To be conducted by TAs.

TAs: Subhranil Deb, Sunaina Pati, Abhishek Goel.

Textbook: ‘Algebra’ by Michael Artin, second edition. Other reference material: ‘Linear Algebra’ by Kenneth Hoffman and Rau Kunze, ‘Linear Algebra Done Right’ by Sheldon Axler.

Course Plan: Throughout the course the emphasis will be on linear algebra over real and complex fields. Topics covered will be - Systems of linear equations, row reduction, vector spaces over any field, bases and dimension, change of basis, linear transformations, dimension formula, matrix of a linear transformation and change of bases, linear operators, similarity, determinants and invertibility, eigenvalues and eigenspaces, characteristic polynomial, triangular and diagonal forms, Euclidean/Hermitian spaces with standard inner product, orthogonal projection, orthonormal bases and Gram-Schmidt procedure, spectral theorem for Hermitian/symmetric operators. The above material corresponds to the following from Artin’s Algebra: chapters 1, 3, 4.1-4.6, parts of chapter 8. If time permits, additional topics will be chosen by the instructor.

Grading: The grade for the class will be decided on the basis of a final exam, one midterm exam, regular in-class quizzes and assignments, and attendance (about 30 percent, 30 percent, 30 percent, and 10 percent respectively). Regular homework assignments will be given, but they are not to be submitted unless I ask you to do so for some of them. But the questions in the quizzes or exams have a high chance of being related to the homework assignments, so it is in your interests to solve homework assignments regularly. Dates of in-class quizzes will not usually be announced in advance. There will be no make-up quizzes for the in-class quizzes and late homework will not be accepted. However, depending upon the number of quizzes we end up having, I will drop for everyone 1 or 2 bottom quiz scores (at my discretion) to compensate for a missed quiz due to necessities and/or emergencies.

Academic Honesty: Given the sterling qualities of character to be found in each and every student at Chennai Mathematical Institute, it is certainly unnecessary to mention that plagiarism and cheating are not only dishonest and immoral, but are also against the policies of CMI. Please keep in mind that, in the highly unlikely event that you do choose to plagiarize or cheat, any such violations will be dealt with in a strict manner in accordance with CMI policies.